Efficient Target Detection of Monostatic/Bistatic SAR Vehicle Small Targets in Ultracomplex Scenes via Lightweight Model

Jiming Lv, Daiyin Zhu, *Member, IEEE*, Zhe Geng, *Member, IEEE*, Hongren Chen, Jiawei Huang, Shilin Niu, Zheng Ye, Tao Zhou, and Peng Zhou

***Abstract*— Military operations often demand considerable concealment and raid capabilities, particularly at night or in adverse weather conditions. However, the use of synthetic aperture radar (SAR) technology provides early warning and target localization capabilities. While spaceborne or airborne SAR systems can capture expansive SAR scenes, they frequently encounter challenges in delivering timely and high-resolution data, thereby limiting their effectiveness in detecting small ground vehicle targets. To address this issue, our research has developed a low-cost, high-resolution, and real-time monostatic MiniSAR system for the effective detection of small targets, such as vehicles. Furthermore, to enhance the stealthiness of the MiniSAR, a bistatic MiniSAR system has been developed to accomplish detection tasks. Nevertheless, despite the utilization of MiniSAR systems for ground armored target detection, two primary challenges persist: the presence of highly ultracomplex scene interference making accurate target detection difficult; and poor real-time performance resulting in slow detection and tracking. To overcome these challenges, this article proposes a ground vehicle target recognition method based on an improved lightweight anchor-free detection network using monostatic/bistatic SAR images. The method initially leverages the inherent features of SAR targets for localization, embedding these features into SAR images, and then outputs detection results through the improved lightweight anchor-free network. We validate the effectiveness of this method on our self-constructed monostatic/bistatic SAR datasets and verify the algorithm's robustness on publicly available ship datasets. Experimental results demonstrate that this method outperforms other representative methods in detecting SAR vehicle small targets, exhibiting higher detection accuracy and timeliness.**

***Index Terms*— MiniSAR, monostatic/bistatic, synthetic aperture radar (SAR) feature localization, vehicle target detection.**

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I. Introduction

THE introduction of unmanned aerial vehicles (UAVs) onto the battlefield is altering the cost dynamics of modern warfare, as low-cost UAVs can trade for numerous high-value targets [1]. In addition, a crucial aspect is that UAVs can rapidly acquire real-time battlefield intelligence, surpassing the speed of satellites. MiniSAR is a system mounted on UAVs that enables ground imaging through synthetic aperture radar (SAR). It boasts advantages such as small size, real-time capability, low cost, and high resolution. Furthermore, bistatic SAR systems offer additional advantages, including providing multiperspective information on targets, enhanced system stealth and anti-jamming capabilities, and greater flexibility in deployment. These attributes provide support for MiniSAR detection of small ground vehicle targets, both in military and civilian applications. However, challenges persist in detection due to imaging system limitations, terrain complexity, and algorithmic considerations, including interference from complex surfaces and limited real-time processing capabilities [2]. Thus, the development of complex ground target detection methods based on monostatic/bistatic SAR data remains a challenging endeavor.

A. Background

Since the rise of deep learning algorithms, significant progress has been made in the field of SAR target detection, such as the saliency-based SAR target detection [3], vehicle detection based on semantic-context enhancement [4], and oriented ship detection in multiresolution SAR imagery [5]. It is vital to note that two challenges still exist. First, complex scenes can interfere with the accuracy of target detection. Second, detection algorithms are often complex, leading to increased time costs for detection.

For the first challenge, due to the complexity of the surface, there are ground scenarios with object interference in SAR images. We define the "ultracomplex scenes" introduced in this article, as characterized by two factors: minimal contrast between the target and background of pixels value, and the target occupies a relatively small portion of the entire image. For instance, as shown in Fig. 1: (a) represents a

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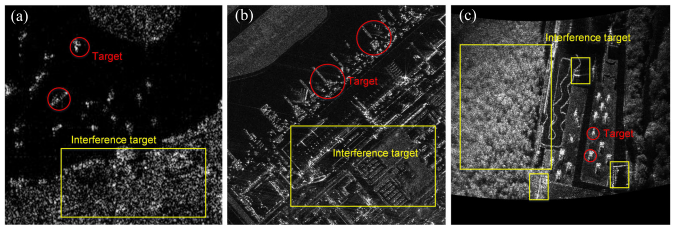


Fig. 1. SAR images scenes of detection: (a) typical scenes, (b) complex scenes, and (c) ultracomplex scenes.

typical scene where the target is clearly separated from the background and occupies a significant portion of the image; (b) represents a complex scene where the background interferes with the target, but the target still occupies a large portion of the image; and (c) represents an ultracomplex scene where the target and the background are interfering with each other, and the small target size make detection challenging. Background interference and target size commonly impact both the accuracy and recall of detection results. Therefore, reducing the clutter interference in the detection scene and accurately obtaining the positional features of the target can facilitate the accuracy of target detection.

The development of SAR image target detection algorithms is divided into two phases separated by the emergence of deep learning [6]. The first stage involves conventional target detection methods, for instance, based on the human visual attention system [7], polarization-based algorithms [8], CFAR [9], and based on wavelet transform [10]. The second stage of SAR target detection methods is implemented through deep learning, such as UIU-Net [11], Faster R-CNN [12], HyperLi-Net [13], and Yolo v7 [14]. Currently, the detection of SAR targets heavily relies on deep learning algorithms [15].

The second challenge lies in the detection algorithms, which can consume a lot of computational time due to more complex images or algorithm inputs. Vehicle targets in ground scenarios differ from those in nearshore scenarios in terms of their relative static state. High-value vehicle targets in ground military scenarios may exhibit dynamic variability. If the detection time is long, the real-time performance of detecting targets will be affected. Therefore, improving the speed of target detection is also a focus of current research.

Enhancing detection speed primarily entails algorithmic innovations. For instance, H2Det [16] adopts fewer convolutional layers, cross-stage partial (CSP) module, and rectangle filling to ensure model high speed. Zhang et al. [13] proposed an improved Yolo V3 to increase the detection speed. Cui et al. [17] proposed a lightweight framework based on a threshold neural network (TNN) to achieve fast detection.

In summary, researchers have recognized that the difficulties in SAR ground-based small target detection stem from complex ground clutter interference and slower detection speeds. They are concurrently striving to address these challenges through advanced small-target localization techniques and accelerated target detection algorithms using SAR imagery.

B. Related Work

1) Improving the Accuracy of Target Detection in SAR Images: Before the beginning of deep learning [18], researchers generally used traditional methods such as CFAR [9], SVM [19], and feature engineering [20] to classify ship and nonship areas with discriminant. Although it can achieve favorable detection results, coarse in many false alarms, the design of feature engineering is complex, and real-time is also worse [15]. Deep learning, an end-to-end approach, overcomes these disadvantages well. For example, YOLO series [21], its variants [22] and SSD [23], a one-step target detection algorithm, can detect the target quickly. Faster-RCNN [24] and RefineDet [25] this two-step detection method, although the detection of the target time is longer, the higher accuracy also makes it receive the attention of researchers. In addition, transformer shows great advantages in target detection, such as detection transformer [26].

The two-step detection algorithm and transformer are superior to the one-step algorithm in terms of detection accuracy but are at a disadvantage in terms of speed. The two-step algorithm is superior in speed but needs to sacrifice detection accuracy. Hence, maintaining a balance between detection accuracy and speed is critical for SAR target detection algorithms. These works suggest that solely relying on deep learning approaches may not be entirely applicable to SAR image target detection; furthermore, it needs to combine the intrinsic properties of SAR imagery, to design methods that can balance accuracy and speed.

2) Improving the Speed of Target Detection Algorithms: Researchers in the domain of SAR image detection are normally more interested in the accuracy and recall of detection and pay relatively less attention to the speed performance of detection algorithms. However, in the area of target detection in computer vision, researchers not only focus on the accuracy of detection but also emphasize the speed of the algorithms [27], [28]. From the perspective of SAR image detection applications, to be able to apply the SAR image detection method to engineering projects, the speed of the approach is an essential part of the research.

In the field of computer vision, Wang et al. [29] proposed a fully learnable group convolution module that is quite efficient and can be embedded into any deep neural network for acceleration. The researcher [30] improved the backbone network of Yolov5s using Mobilenetv3-small, which reduces the number of model parameters and computations and increases the network speed. Sommer et al. [31] proposed an adaptive model that reduces the input area of the detection module and thus decreases the inference time. While researchers in the field of computer vision are predominantly focused on this issue, there is also a subset of researchers in the SAR image detection domain who are paying attention to this problem. Shipdenet-18 [32] is a 1 Mb lightweight deep learning network with only 18 convolutional layers, which can effectively improve the detection speed. A novel lightweight basic network with feature relay amplification and multiscale feature hopping connection structure to enhance detection speed and accuracy [33]. Ma et al. [34] proposed a lightweight detector Light-YOLOv4, which is obtained by model compression of YOLOv4, and the detection speed is enhanced by 2.4 times compared to the original YOLOV4.

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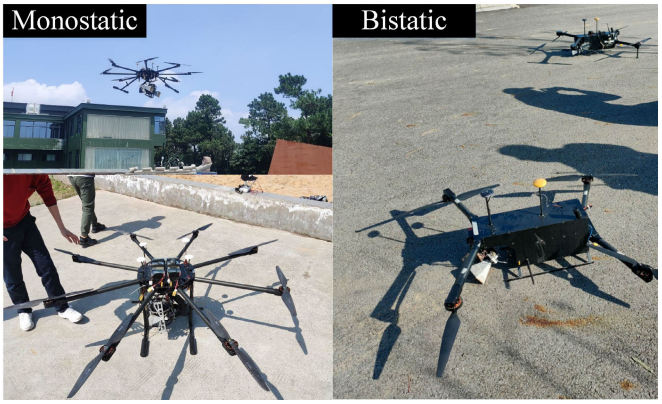


Fig. 2. Photographs of the MiniSAR system of NUAA.

TABLE I
ABBREVIATIONS AND ENTIRE NAME MAPPING OF TARGET TYPES

Index	Abbr.	Entire name
1	62LT	Domestic Type 62 Light Tank
2	63APTV	63 Armored Personnel Transport Vehicle
3	63CAAV	63C Amphibious Armored Vehicle
4	63AT	Domestic 63A Amphibious Tank
5	T34-85	Soviet T34-85 Medium Tank
6	59-1TC	Domestic 59-1 130mm Tower Cannon
7	54TH	Homemade 54 122mm Towed Howitzer
8	59AG	Domestic 59 Anti-aircraft Guns
9	J6	Shenyang J-6
10	Radar	Ground-based detection Radar
11	AMG	Antiaircraft Machine Gun

In conclusion, from these related works, we can realize that enhancing the detection speed is an essential process for applying SAR target detection to engineers. Therefore, from the perspective of model structure and model requirements, our method aims to reduce the model size and increase the detection speed.

C. Main Contribution and Organization

To address these two challenges and enhance the practicality of utilizing SAR for detecting small ground vehicle targets. Researchers need to develop a new approach to deal with the interference of ultracomplex ground clutter and the problem of algorithm timeliness. Therefore, this article proposed a deep learning processing framework named location tiny YoloX (LTY-Network). We conducted a comparative verification method and passed analysis was performed using publicly available detection datasets and self-developed monostat/bistatic SAR datasets, and the accuracy and speed of the model were evaluated. Furthermore, we conducted ablation experiments and compared the proposed SAR target detection algorithm with representative algorithms, further analyzing the superiority of the algorithm. The main contributions include the following.

1) A novel LTY-Network is proposed, which is based on SAR image feature localization and an improved lightweight anchor-free detection network. An end-to-end deep learning network is constructed to address the challenge of detecting highly complex ground small vehicle targets. On the full aspect stationary targets-vehicle (FAST-Vehicle) and ship datasets, the proposed network exhibits higher detection accuracy and speed compared to existing methods.
2) To verify the effectiveness of the algorithm, we extend the bistatic SAR dataset based on the FAST-Vehicle monostatic SAR dataset by carrying out multiple flights to capture the targets. The dataset is more abundant in bistatic depression angle and bistatic azimuth angle information, which effectively expands the dataset of SAR ground small vehicle targets. Furthermore, it was demonstrated that the bistatic SAR images can also detect ultracomplex ground small vehicle targets.
3) This article compares and validates the proposed method via various SAR target datasets. We analyze the impact of our method on experimental results under distinct conditions as well. In addition, the precision and efficiency of the proposed method are compared with other state-of-the-art target detection algorithms.

The article is organized as follows. Section II describes the monostatic and bistatic SAR datasets and shows sample SAR images. Section III proposes an LTY-Network algorithm to detect SAR small target images on the complex ground. Section IV shows the experimental results of the proposed algorithm for various SAR image datasets, respectively. Finally, Section V summarizes the entire article.

II. DATASETS DESCRIPTION

A. MiniSAR Overview

The MiniSAR system possesses the advantages of lightweight, cost-effectiveness, and real-time capability, enabling target detection in regions of interest without illumination or shooting time limitation. In this project, the research team has developed a monostatic and bistatic SAR dataset using the MiniSAR system (see Fig. 2). This work represents the final extension within the project focused on SAR target detection and recognition. Based on the previous two iterations involving monostatic data [35], [36], along with the current work utilizing bistatic data, the three works constitute a complete system of MiniSAR target recognition and detection. FAST-Vehicle encompasses a total of 11 target types (see Table I), with optical images for each target provided in Fig. 3. Moreover, this article also shows a monostatic/bistatic SAR panoramic image.

B. Monostatic MiniSAR

The main parameters of monostatic and bistatic MiniSAR systems are listed in Table II. Specifically, the MiniSAR system operates in the X-band and spotlight mode, with a bandwidth of 1800 MHz, achieving approximately 0.1-m range resolution to ensure submeter-level image details. The system is configured with a Doppler frequency modulation slope of 9×10^{13} and a pulsewidth of 2 ms. In addition, by controlling the flight altitude and distance, the radius of the flight trajectory is approximately 350 ($\pm$5) m. The length of

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TABLE II
CORE PARAMETERS OF MINISAR

System parameters	Monostatic Depression					Bistatic	Unit
	15°	26°	31°	37°	45°		
Mode			Spotlight				-
Center frequency			9.7				GHz
Sampling rate			100				MHz
Bandwidth			1800				MHz
Pulsewidth			2			1.9012	MS
PRF			500			525	Hz
Platform height	94	150	210	265	350	-	M
Operation distance	360	335	407	439	495	-	M
Platform speed			5-7				M/S
Resolution			0.1*0.1			0.16*0.26	M*M



Fig. 3. Optical image of the target.

one revolution trajectory is around 2200 m, with a flight time of approximately 360 s. We set the pulse repetition frequency (PRF) of the system to 500 Hz and azimuth sampling pulses to 4096, with data multiplexing applied. According to theoretical calculations, data can be acquired every 1.2 s, meaning the system can capture 300 panoramic images during one flight.

C. Bistatic MiniSAR

Since the bistatic MiniSAR receiver and transmitter are not on the same UAVs (see Fig. 4), there are some differences in the parameters between the bistatic and monostatic configurations (see Table II). These differences include pulsewidth a pulsewidth of 1.9012 ms, a PRF of 525 Hz, and a resolution of 0.16×0.26 m. To acquire more bistatic SAR images, a total of nine flight missions were conducted. The specific information regarding azimuth and depression angles for each mission can be found in Table III. Furthermore, in terms of target type, we added a class of target AMG to the FAST-Vehicle. The bistatic SAR images were taken at a relatively close time, both in November 2023. Due to the distinctness in the geometry of signal transmission and reception between bistatic SAR and monostatic SAR, there are discrepant features in their images. Through observing Fig. 5, the separation between the transmitter and receiver can result in diverse resolutions in the azimuth and range. The differing geometry may also cause geometric distortions in the images. Moreover, the various signal paths may cause variations in shadow length and target contrast in bistatic SAR, depending on the angle between the transmitter and receiver. In bistatic SAR, because of the more complex signal paths and geometry, the image contrast and speckle may vary. This variation can sometimes enhance certain features but may also lead to uneven image quality. Although monostatic SAR generally produces higher image quality, bistatic SAR provides more information and views.

III. METHODOLOGY AND PROCESS

A. Overview of Framework Structures

To improve the accuracy and speed of small target detection in complex ground scenes with MiniSAR, this article proposes an LTY-Network method. The method consists of two ideas. The first one is to locate the target utilizing the intrinsic properties of the SAR image and assign the localization features; the second one is to improve the YoloX to obtain a more lightweight network. In addition, to explore the usefulness of our method, we give the training process and analysis. By integrating these two enhancements, the proposed detection method can be effectively applied to the SAR detection of ultracomplex ground small vehicle targets.

B. Target Localization Method Based on SAR Image Features

In recent years, SAR image target detection has relied on computer vision-based methods, which primarily focus on optical images, but SAR targets differ significantly from human visual perception. Therefore, for SAR image interpretation, it is necessary to develop new target detection methods that take into account the characteristics of SAR images. Addressing this requirement, this article constructs new SAR image target localization features to assist target detection. The SAR target localization method consists of two steps: the first step is target localization according to scattering characteristics and texture features, while the second step involves region-growing target localization based on the probability density distribution of tiled pixels. Furthermore, with the results of these two steps, this method exploits different intersections over union (IoU) of the same target to filter each other and finally achieve the target location information. We show the flowchart of the method (see Fig. 6).

1) Target Localization Based on Scattering Center and Texture Features: In SAR target detection research, the scattering center and texture features of SAR images are commonly utilized. Consequently, we have developed a new method for target localization based on the scattering center and texture features. The purpose of this method is to attain the possible locations A of targets in SAR images

$$I_t = \max\left(f_{\text{gabor}}\left(\left[\text{Img}_{c1}, \text{Img}_{c2}, \text{Img}_{c3}\right], \text{Ks} = 3 \right.\right. \\ \left.\left. \text{Dir} = 8, \text{Freq} = [0.2, 0.4, 0.6]\right)\right) \quad (1)$$

where Ks denotes the kernel size, Dir is the direction, and Freq is the for frequency.

Specifically, this method first extracts the texture feature map I_t of the input SAR panorama image Img using the Gabor filter as shown in Fig. 7(a) with (1)

$$I_{\text{sc}} = f_{\text{filter}}(I_t, \text{Ks} = 4). \quad (2)$$

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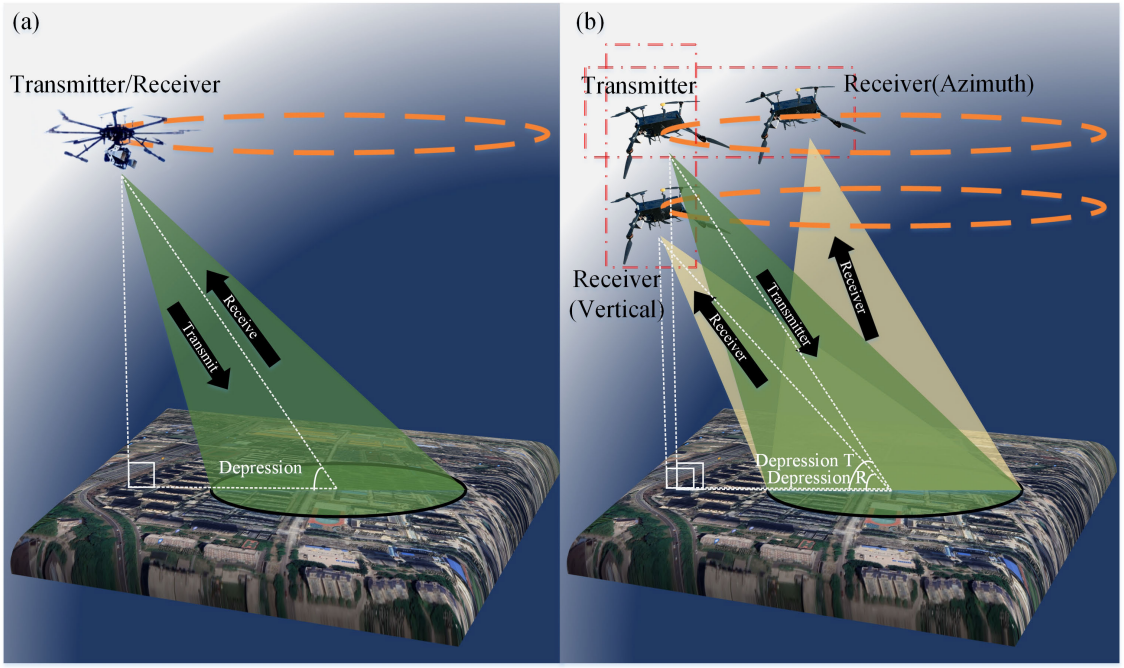


Fig. 4. Monostatic and bistatic MiniSAR imaging simulation: (a) monostatic imaging geometry simulation and (b) bistatic imaging geometry simulation.

TABLE III
TRANSMITTER AND RECEIVER ANGLE INFORMATION FOR DIVERSE FLIGHTS OF BISTATIC MINISAR

Flight	Transmitter depression	Receiver depression	Azimuth bistatic angle
1	15°	45°	0°
2	15°	60°	0°
3	30°	45°	0°
4	40°	45°	0°
5	45°	45°	5°
6	45°	45°	15°
7	45°	45°	30°
8	45°	45°	45°
9	30°	30°	30°

Subsequently, a window with a core size of (4, 4) [see (2)] is utilized to traverse the image and gain the image scatter center I_{sc}, illustrated in Fig. 7(b)

$$I_{t_\text{sc}} = f_{\text{replace}}(I_{\text{sc}}(x, y), I_t(x, y)). \tag{3}$$

To integrate the scattering center features and texture features, I_{sc} is set as the reference image, and image extraction is performed on I_t, so that the scattering center pixels of I_{sc} are replaced with texture pixels of I_t to acquire the image I_{t_sc} [see (3)]

$$I_{\text{sparse}}(x, y) = f_{\text{erdoe}}\left(f_{\text{dilate}}\left(I_{t_\text{sc}}(x, y)\right)\right). \tag{4}$$

Next, dilation and erosion operations are applied to I_{t_sc} [see (4)] to acquire sparse SAR images with low noise, as depicted in Fig. 7(c).

Although SAR feature images of I_{sparse} were available, they are insufficient to determine the possible location A definitively. As can be seen from the zoomed-in view image in Fig. 7(c), the vehicle targets 59AG and T34-85 in the sparse map I_{t_sc} are blurred. To facilitate the scanning of the targets

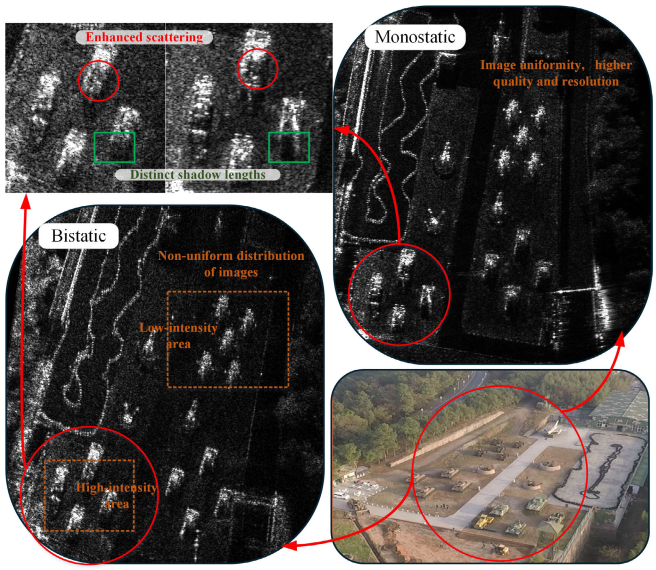


Fig. 5. Sample images of monostatic and bistatic SAR imaging scenes.

in the following steps, the contours of the sparse targets 59AG and T34-85 should be strengthened:

$$I_{\text{fill}}(x, y) = f_{\text{fill}}\left(f_{\text{dilate}}\left(I_{\text{sparse}}(x, y)\right)\right). \tag{5}$$

For this purpose, the article undertook a two-step approach [see (5)]. First, the method dilated I_{sparse} to obtain contour boxes for all block and point regions. Second, we reinforced the features by filling in all the pixels within these contours, resulting in the image I_{fill} [see Fig. 7(d)]

$$\text{Possible location } A = f_{\text{laplace}}\left(I_{\text{fill}}(x, y)\right)) * \text{factor}. \tag{6}$$

The figure I_{fill} shows the initial location of the target, but it still failed to ascertain the more accurate location of the

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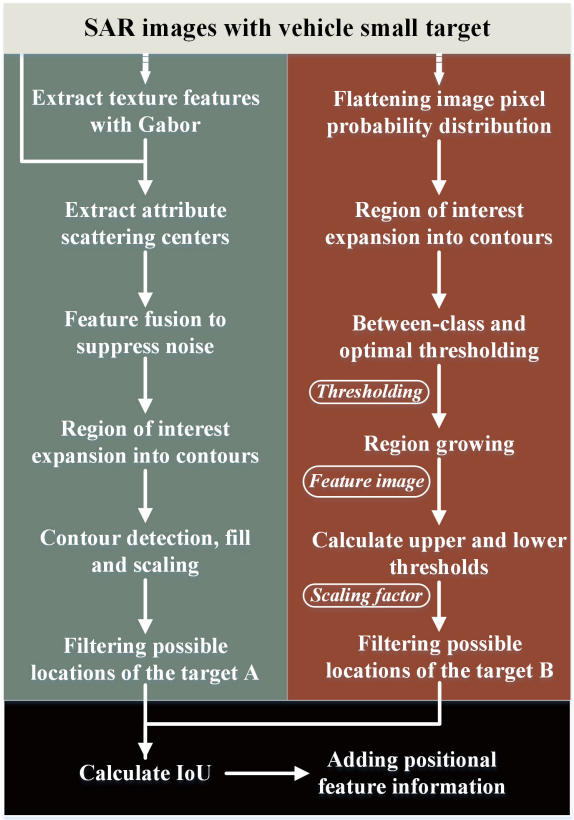


Fig. 6. Framework of the proposed target localization method.

target. Hence, the next step is to determine the target location as well. Herein, the Laplacian algorithm is employed to extract the edge information from I_{fill}. Within the edge contour of I_{fill}, to make the sparse pixels within the contour clustered together, we additionally set a scaling factor factor with a value of 1.2. As a result, the first location constraint map of the possible location A can be acquired [see Fig. 7(e)].

2) Target Localization for Region Growing Based on Uniform Image Probability Density Distribution: We computed the probability density function (pdf) of all the SAR images in the FAST-Vehicle dataset (see Fig. 8(b) shows sample images). It revealed that the grayscale distribution of our dataset is not uniform, which leads to the loss of detailed information in the images and reduces the informativeness of the images. Hence, a new method is proposed to achieve a uniform probability density distribution for SAR images, resulting in Fig. 8(c), termed the equivalent interpolation uniform probability density (EIUPD) method.

The specific method is: from Fig. 8(b), it is evident that most pixel values in the original pdf are absent. It is because the number of image elements is too few resulting in not visualizing the bars in Fig. 8(b). To mitigate computational complexity (CC) and balance the number of pixels, a threshold is set based on the following equation to eliminate these infrequent pixel values:

$$\text{Threshold}_{\text{pdf}} = f_{\text{mean}}(\text{pixel num per}) * 0.05 \tag{7}$$

where f_{mean} denotes the mean value, pixel num per represents the number of pixels value, and setting the total number of pixels to 0.05 essentially removes the infrequent elements. Based on extensive experimental data, setting the threshold at 0.05 effectively removes invalid pixel values. When the threshold is set to this value, invalid gray values can be excluded to prevent interference with the subsequent steps of obtaining the equivalent distance. In other words, gray values with a higher proportion need to be preserved, while the sparse, invalid gray values should be maximally suppressed.

After this processing step, a list is formed that contains all pixel values exceeding the $\text{Threshold}_{\text{pdf}}$, denoted as $\text{list}_{\text{value}} = [\text{value EIU}_0, \text{value EIU}_1, \ldots, \text{value EIU}_n]$. Subsequently, the gap between each pixel value and the previous pixel value is calculated according to the following equation:

$$\text{gap}_{n-1} = \begin{cases} \text{value EIU}_1 - \text{value EIU}_0, & n = 1 \\ \text{value EIU}_n - \text{value EIU}_{n-1}, & \text{other.} \end{cases} \tag{8}$$

A new list is systematically constructed $\text{list}_{\text{num}} = [\text{num EIU}_0, \text{num EIU}_2, \ldots, \text{num EIU}_n]$, by associating the pixel values in the $\text{list}_{\text{value}}$ with their corresponding pixel counts. The values of the list_{num} are then evenly divided into gap partitions according to (9). Furthermore, a special case needs to be considered where $\text{num EIU}_1 = \min$. Since SAR images tend to have numerous dark pixels, resulting in min being equal to 0. When $\text{value EIU}_n = \text{value EIU}_0$, there is $\text{gap} = \text{num EIU}_1 - \text{num EIU}_0$

$$\text{num EIU} = \begin{cases} \dfrac{\text{num EIU}_n}{\text{gap}_0}, & n = 1 \\ \dfrac{\text{num EIU}_n}{\text{gap}_{n-1}}, & \text{other.} \end{cases} \tag{9}$$

The new uniform pdf in Fig. 8(c) is achieved by this method. Then, depending on this pdf, the number of pixel values in the original SAR image was adjusted to make the gray scale distribution of the SAR image more uniform [see Fig. 8(d)].

In accurately locating the targets via region-growing methods, it is crucial to set reasonable segmentation thresholds. Upon the foundation of a uniform pdf for the SAR image by EIUPD, the optimal threshold can be achieved by employing the Otsu method based on interclass variance. It is worth noting that, the original SAR image had a severely discrete grayscale distribution, whereas the image processed by the EIUPD method shows a more uniform grayscale distribution. Consequently, the Otsu threshold values between the two may differ significantly. Images with highly discrete pixel distributions tend to produce an Otsu threshold value that skews toward the extremes, while those with more uniform grayscale distributions tend to generate a threshold closer to the midpoint. This is because the Otsu method determines the optimal threshold by maximizing the interclass variance, and the characteristics of the grayscale distribution have a significant impact on this calculation. Generally, if the pixel values are concentrated in the low grayscale range, the Otsu threshold tends to be lower, while the concentration in the higher grayscale range results in a higher threshold. It is evident that the pixel values of SAR images are concentrated within a specific range. Without using the EIUPD method, the resulting image segmentation might obscure the target,

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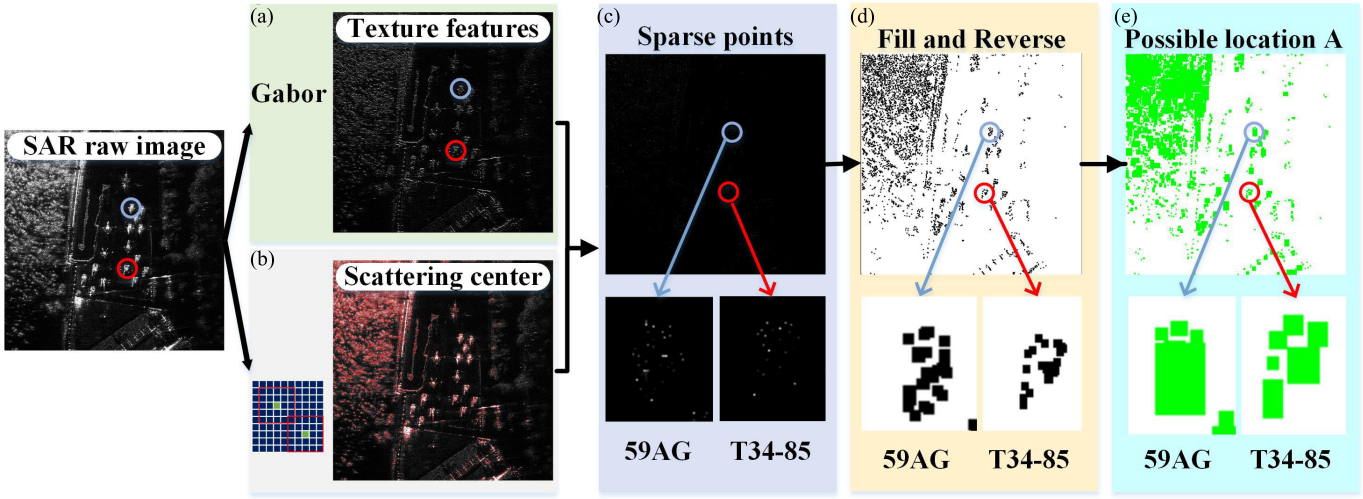


Fig. 7. Example of target localization method with SAR image features: (a) texture feature map, (b) scattering center map, (c) extract sparse points, (d) fill and reverse, and (e) determine possible location A.

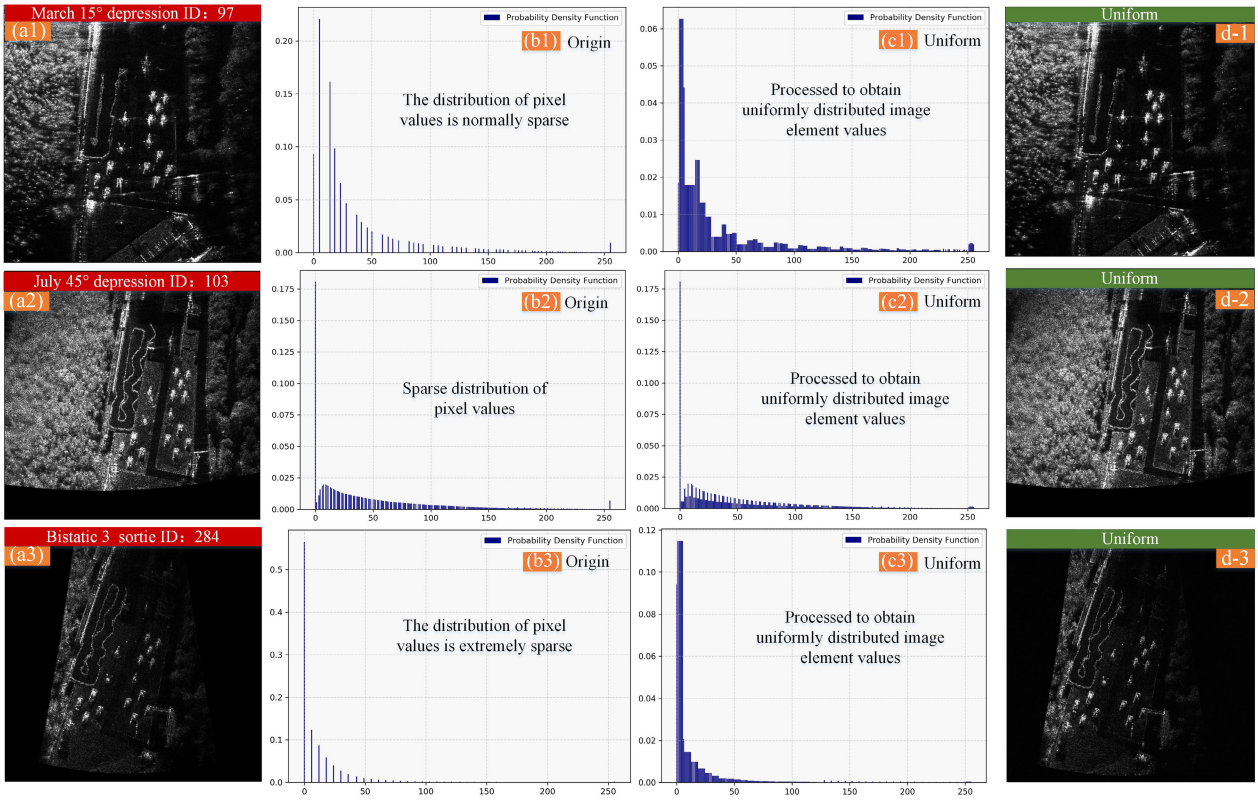


Fig. 8. Sample SAR images processed by EIUPD: (a-1)–(a-3) raw SAR images, (b-1)–(b-3) pdf of raw images, (c-1)–(c-3) pdf with EIUPD, and (d-1)–(d-3) images with EIUPD.

whereas the processed image more effectively distinguishes between background and noise.

The exact method can be known from the Algorithm 1: first, an array of the same size as the input image is created to store the output segmentation results. The array is initialized to all zeros, indicating that all pixel points have not been accessed. In addition, eight connectivity directions are defined, including up, down, left, right, and the four diagonal directions. These directions are used to determine the neighboring pixels of the current pixel [see Fig. 9(c)]. The selection of the seed point is an essential step, which is a point belonging to a known target region. Nevertheless, since the background of the task is an unknown target region, this method elects five seed points whose centers cover the whole image, including the center point of the image and the midpoints of the four diagonals as starting points [see Fig. 9(c)]. Furthermore, a queue is created, and seed points are added to the queue. The queue is used to store pixels awaiting processing. The vital step is to perform region growing on the SAR image. Constantly take pixel points from the queue and mark them as target regions.

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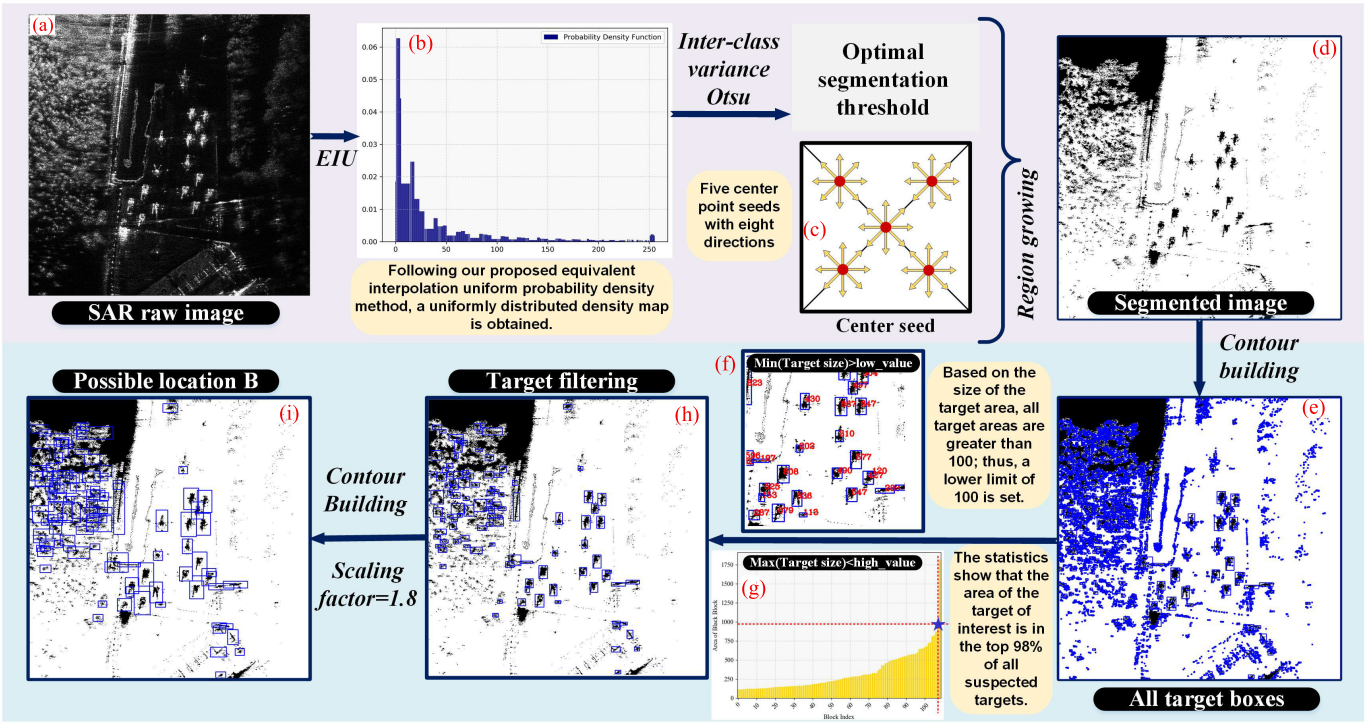


Fig. 9. Process of target localization method based on improved region growing: (a) raw image, (b) pdf, (c) five center point seeds, (d) segmented image, (e) targets in boxes, (f) limiting the target's size, (g) size thresholds, (h) filtering, and (i) possible location B.

Algorithm 1 Improved Region-Growing Segmentation

```
Initialize segmentation result image I_rg with all zeros
Define seed points set S_five
Define queue Q
for s ∈ S_five do
    Add seed point s to queue Q
end for
while Q is not empty do
    Remove pixel p from queue Q
    if p is not visited then
        Mark p as accessed
        for each neighbor pixel p_neighbor of p do
            if p_neighbor is not accessed and is similar to p then
                Add p_neighbor to queue Q
                Mark p_neighbor as part of target region
            end if
        end for
    end if
end while
Perform edge detection on segmentation result image I_rg to obtain contours
Set threshold values T_1 and T_2 for suppressing redundant information
```

The eight neighboring pixels of the current pixel point are checked and if the neighboring pixel satisfies the condition that the current image element value is to be less than the optimal threshold (not accessed and similar to the seed pixel), it is inserted into the queue and marked as the target region. This process continues until the queue is empty and stops. Finally, it is capable of gaining the image I_{rg} after region-growing segmentation.

However, obtaining I_{rg} alone does not accurately determine the position of the targets, as the image still contains redundant information. Thus, further processing is imperative. In detail, the approach involves first obtaining all contour information of I_{rg} through contour detection. It is worth mentioning that, because of the more scattered I_{rg} and the existence of a large number of noise points, we improved this step to suppress the useless information. Through Fig. 9(f) and (g), it is found that the pixel size of the target of interest is generally larger than 100 and the area is less than 0.98 of the maximum pixel value. As a result, two upper and lower thresholds T_1 and T_2 are set to suppress the redundant information and to further locate the second position possible location B

$$\text{IoU}_{AB} = \frac{S_A \cap S_A}{S_A \cup S_A}. \tag{10}$$

After the above two steps, possible location A and possible location B are attained. To pinpoint the location of the target of interest, this study designs an IoU filtering measure that calculates the overlap area of possible location A and possible location B [see (10)]. When the IoU value reaches 0.1, the generating frame of the target area is retained and the rest of the generating frames are deleted. The selection of the IoU threshold in this work is based on empirical data derived from the images. For both monostatic SAR data and bistatic SAR data, when the IoU is set to 0, a significant amount of noise remains unfiltered. However, when IoU exceeds 0.1, many targets are missed. Therefore, 0.1 is the best alternative in this step. Moreover, a scaling factor of 1.8 was assigned to cover the target as much as possible. In the end, this method

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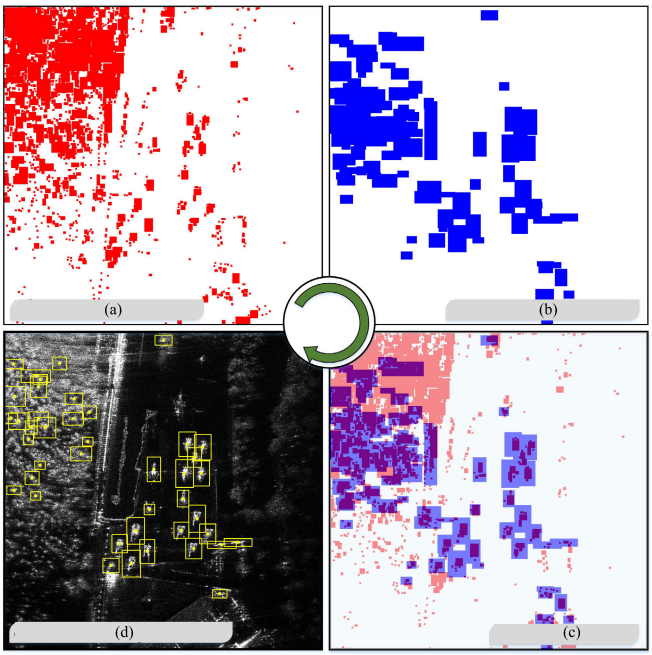


Fig. 10. Process of IoU filtering for target location: (a) possible location A, (b) possible location B, (c) position A and B overlap, and (d) final result.

yields the final result of locking the location localization of the target of interest after SAR image intrinsic feature filtering (see Fig. 10).

C. Fast Anchor-Free Frame Detection Algorithm

To reduce the model size of the SAR target detection algorithm, while maintaining detection accuracy to a reasonable extent, we utilize YoloX-S as the base algorithm [37], which balances accuracy and speed. To achieve this goal, the study proposes two modifications to the base algorithm. First, leveraging the enhanced depthwise separable convolution (DSC) technique, as referenced in MobileNet [38], decomposes standard convolutions into depthwise convolutions and pointwise convolutions to decrease computational and parameter complexity. Second, for tasks involving the detection of small objects, simplifying the detection head of the base method designed for detecting large objects can reduce the network layers. Fig. 11 illustrates the designed network framework, which is divided into two parts: the upper part presents specific module details, while the lower part depicts the overall architecture of the network.

1) DSC Based on Spatial Attention Mechanism Substitutes the Standard Convolutional: As a general rule, convolution in YoloX networks is mostly applied in convolution blocks, Resnet blocks, spatial pyramid pooling (SPP) resnet blocks, and CSP. With standard convolution operations performed simultaneously on all input channels, a large number of convolutional layers can lead to a delay in the network operation (see Fig. 12). To compare the operational efficiency of a standard convolutional neural network (CNN) with the improved DSC based on spatial attention mechanism, we evaluate their CC. For a standard CNN, assuming the size of the input images in Fig. 12 is $D_w \times D_h \times N_c$, and the size of the kernel size is $D_k \times D_k \times M_c$. Furthermore, the CNN generates the feature maps for each input channel of each input image through the convolutional kernel.

Thus, the computation of a single convolutional kernel is

$$\mathrm{CC}_{\mathrm{single}} = D_w \times D_h \times D_k \times D_k \times N_c. \tag{11}$$

Furthermore, the computation of the extrapolated N convolutional kernels is

$$\mathrm{CC}_{\mathrm{sc}} = \mathrm{CC}_{n_k} = D_w \times D_h \times D_k \times D_k \times N_c \times M_c. \tag{12}$$

While the YoloX-Nano of the developers exhibits higher operational efficiency, it decreases interchannel information exchange, thereby limiting the expressive abilities of the model and reducing detection accuracy [37]. Hence, the first improvement scheme is for the convolutional layer widely used in YoloX-S. Attention DSC (ADSC) based on the attention mechanism is utilized to decompose the standard convolution in YoloX-S into depth convolution and point-by-point convolution with a spatial attention mechanism to enhance the information sharing between different channels.

Deep separable convolution separates the standard convolution operation into two steps: first, an individual convolution operation is performed on each input channel (deep convolution), and then the output channels of the deep convolution are combined at each position (point-wise convolution) based on a spatial attention mechanism. Utilizing spatial attention mechanisms before point-wise convolutions enables the model to focus on various regions of the image to better learn the semantics of the image while executing the task. As shown in Fig. 13, the spatial attention mechanism primarily consists of three components: max pooling and average pooling, convolution, and an activation function. First, the input feature map is processed through both average pooling and max pooling, producing two distinct feature maps. These pooled feature maps are then concatenated along the channel dimension and passed through a convolutional layer to obtain the spatial attention feature map. The Sigmoid function [see (13)] is applied to constrain the values of the attention map between 0 and 1. Finally, the original input feature map is weighted accordingly. Utilizing spatial attention mechanisms before point-wise convolutions enables the model to focus on various regions of the image to better learn the semantics of the image while executing the task

$$F' = \sigma(\mathcal{W}_a * F) \odot F. \tag{13}$$

$\mathcal{W}_a$ is the weight of the convolution operation, F is the input feature map; * denotes the convolution operation, σ is the Sigmoid function, and $\odot$ denotes the element-by-element multiplication.

Point-wise convolutions enhance the representation capability of input features, enhancing the ability of the network to perceive the target, which in turn results in increased performance in detection or classification. Finally, putting the spatial attention mechanism after the deep convolution allows the network to extract more advanced features. The features are then weighted and fused according to the global or local level of attention to further improve the accuracy and robustness of the feature representation. Decomposing standard convolutions

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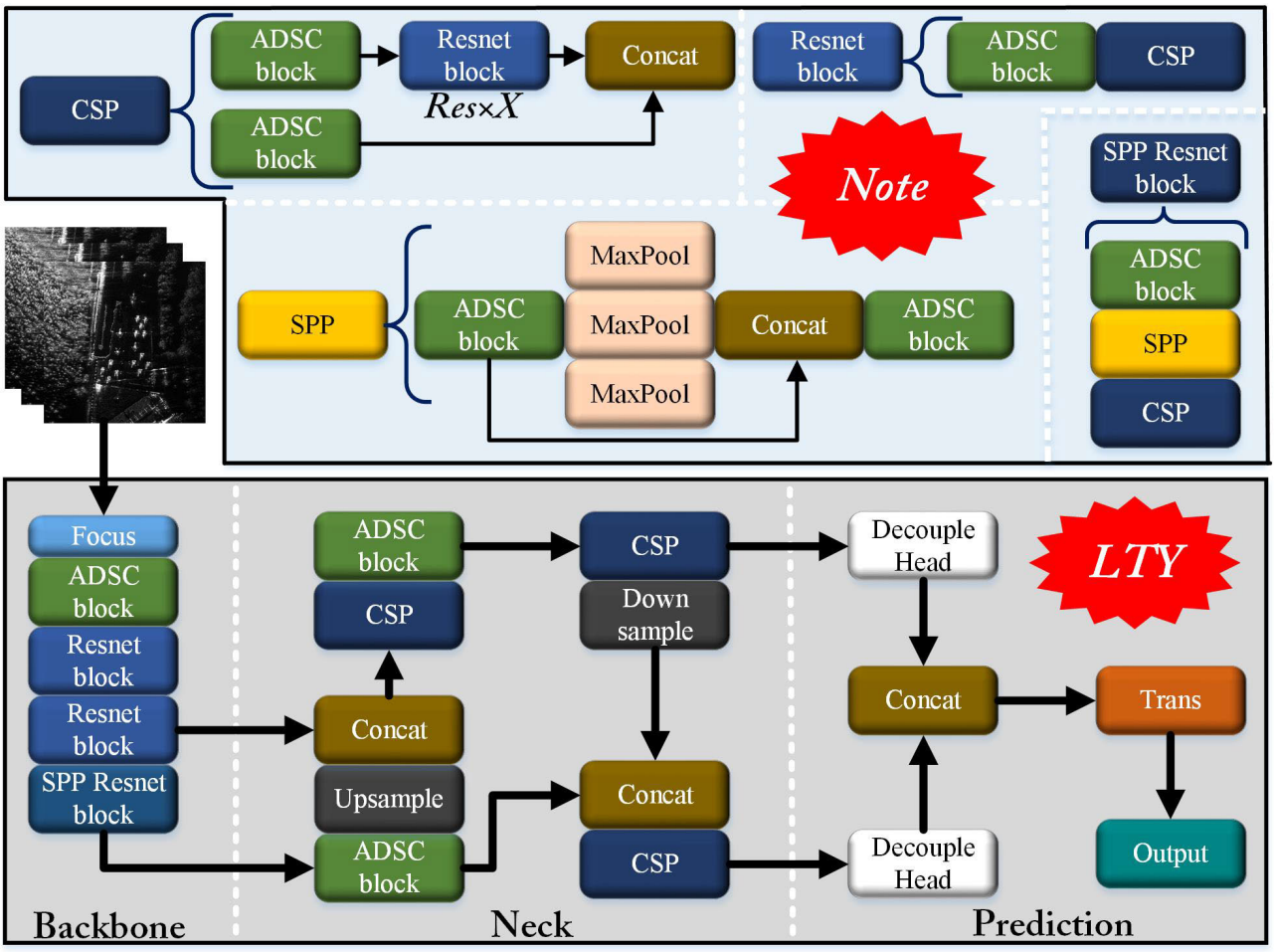


Fig. 11. Improved lightweight anchor-free frame detection network framework.

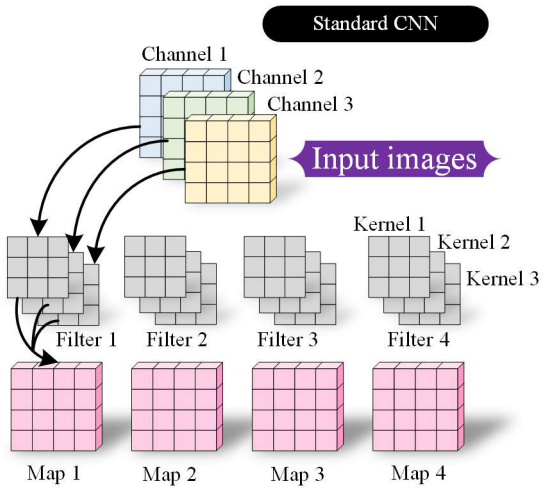


Fig. 12. Standard convolution operation principle.

into ADSC can reduce computational and parameter costs, thereby improving the efficiency of target detection algorithms. The specific network structure is shown in Fig. 13. Similarly, we also need to compute the computational cost of ADSC.

To begin with, the CC of channel-wise convolution is calculated as follows:

$$\mathrm{CC}_{\mathrm{dc}} = D_w \times D_h \times D_k \times D_k \times N_c. \tag{14}$$

Then, the computational amount of the spatial attention mechanism is typically small, and its computational amount mainly comes from the size of the feature map, the number of channels, and the parameters of the convolutional layers employed in the attention mechanism. The computational effort is low compared to a large number of convolutional operations

$$\mathrm{CC}_{\mathrm{attention}} = f(\mathrm{head}, D_{\mathrm{key}}). \tag{15}$$

At last, the CC of pointwise convolution is

$$\mathrm{CC}_{\mathrm{dp}} = D_w \times D_h \times N_c \times M_c. \tag{16}$$

Finally, the total computation of ADSC is the sum of the above three

$$\mathrm{CC}_{\mathrm{ADSC}} = \mathrm{CC}_{\mathrm{dc}} + \mathrm{CC}_{\mathrm{attention}} + \mathrm{CC}_{\mathrm{dp}}. \tag{17}$$

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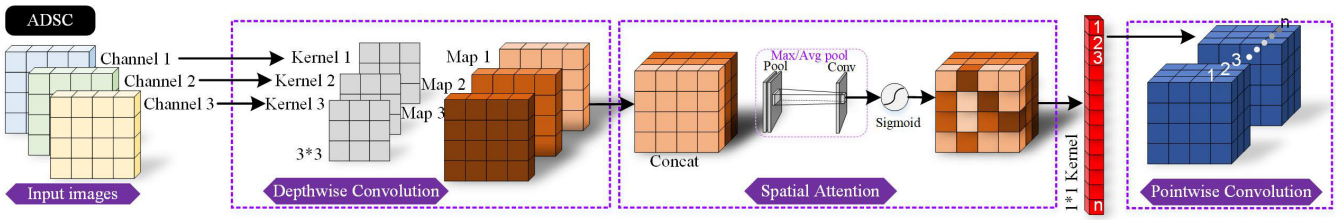


Fig. 13. Network structure of ADSC.

In summary, this study compares the computational complexities of standard convolution and ADSC

$$\frac{\mathrm{CC_{ADSC}}}{\mathrm{CC_{sc}}} = \frac{\mathrm{CC_{dc}} + \mathrm{CC_{attention}} + \mathrm{CC_{dp}}}{D_w \times D_h \times D_k \times D_k \times N_c \times M_c} = \frac{1}{M_c} + \frac{1}{D_k^2} + \frac{D_{\mathrm{key}}}{D_w \times D_h \times D_k^2 \times N_c \times M_c} + \frac{\mathrm{head}}{D_k^2 \times M_c}. \tag{18}$$

Furthermore, due to D_{key} and the head exhibit orders of magnitude difference compared to the denominator, the third and fourth terms of (18) can be neglected. This approximation yields

$$\frac{\mathrm{CC_{ADSC}}}{\mathrm{CC_{sc}}} \approx \frac{1}{M_c} + \frac{1}{D_k^2}. \tag{19}$$

Assuming, the convolution kernel is a regular 3×3, then $D_k = 3$. In other words, the computation of $\mathrm{CC_{ADSC}}$ is only 1/9 of that of $\mathrm{CC_{SC}}$, in the case that M_c is not particularly small. Therefore, it can be concluded that our proposed ADSC can greatly reduce the number of parameters of the model and thereby enhance the computational efficiency.

2) Simplified Large Target Detection Head: To enhance the speed and efficiency of vehicle small target detection in SAR images, this study introduces modifications to the "Backbone" structure of the YoloX-S network. The original YoloX-S network contains three detection heads for detecting small, medium, and large objects. However, since the focus of this work is on vehicle target detection, the detection head for large objects is removed to simplify the network. To ensure the effectiveness of this modification, adjustments were made not only to the "Prediction" part but also propagated from the "Prediction" to the "Neck" and further to the "Backbone." Fig. 14 illustrates the modified network structure.

The backbone of YOLOX-S is based on CSP-Darknet53, which initially utilizes a "Focus" structure to reduce information loss, followed by two ADSC blocks to extract target features. This study modifies the network by removing the middle detection layer, improving the resolution of the final feature map. The SPP module is retained to enhance the adaptability and receptive field of the model, enabling better handling of targets at different scales while maintaining high efficiency. This improves both the accuracy and robustness of target detection. As a result, the modified backbone generates the fundamental features feature base small (FBS) and feature base middle (FBM).

In the neck of the network, the feature fusion of YoloX-S employs a combination of feature pyramid networks (FPNs) [39] and path aggregation network (PAN) [40], facilitating bidirectional feature fusion from top to bottom and bottom to top. Initially, FBM obtained from the backbone is processed via ADSC to yield the small target feature layer feature neck middle 1 ($\mathrm{FNM_1}$). $\mathrm{FNM_1}$ is then concatenated with $\mathrm{FNM_2}$, derived from the small target feature layer, and inputted into the CSP layer to obtain new $\mathrm{FNM_2}$. Moreover, for the features of the small target feature layer, $\mathrm{FNM_2}$, upsampling of $\mathrm{FNM_1}$ is conducted and concatenated with FBS. Then, after the $\mathrm{FNS_1}$ obtained by the CSP layer and the ADSC module, another layer of CSP is used to obtain the feature $\mathrm{FNS_2}$ of the final neck. This approach effectively streamlines the network structure while maintaining minimal loss in accuracy, tailored for small target detection tasks.

In the prediction section, the original network features three decoupled heads utilizing feature layers F_{L1}, F_{M2}, and F_{S3} of large, medium, and small scales, respectively, for predicting bounding boxes and conducting loss calculations. Here, building upon the output of the neck section, two decoupled heads, the decouple head 1 and the decouple head 2, corresponding to medium and small target detection heads, are introduced. Concretely, category scores are obtained via two ADSC modules, while regression scores (Reg) and object existence scores (Obj) are gained. Ultimately, these modifications complete the network transformation.

In conclusion, this study enhances the practicality and efficiency of the SAR target detection algorithm through two critical improvements. First, the YoloX-S network structure is simplified by removing the large target detection head, which is particularly suitable for small target detection scenarios. Second, the standard convolution is decomposed into depth convolution and point-wise convolution by utilizing DSC, which effectively reduces the amount of computation and the number of parameters. These enhancements provide crucial technical support for advancing target detection algorithm performance and applications.

D. Relationship Between Location and Detection Algorithms

The method proposed in this article maintains high detection accuracy while improving the detection speed, depending on the target localization information. In Section III-B, the preliminary position of the object is determined based on the inherent features of the target in SAR images. This position is referred to as "Location_A." In addition, an improved image segmentation method is employed to obtain another possible position of the target, called "Location_B." By cross-referencing these two location estimates, a more accurate

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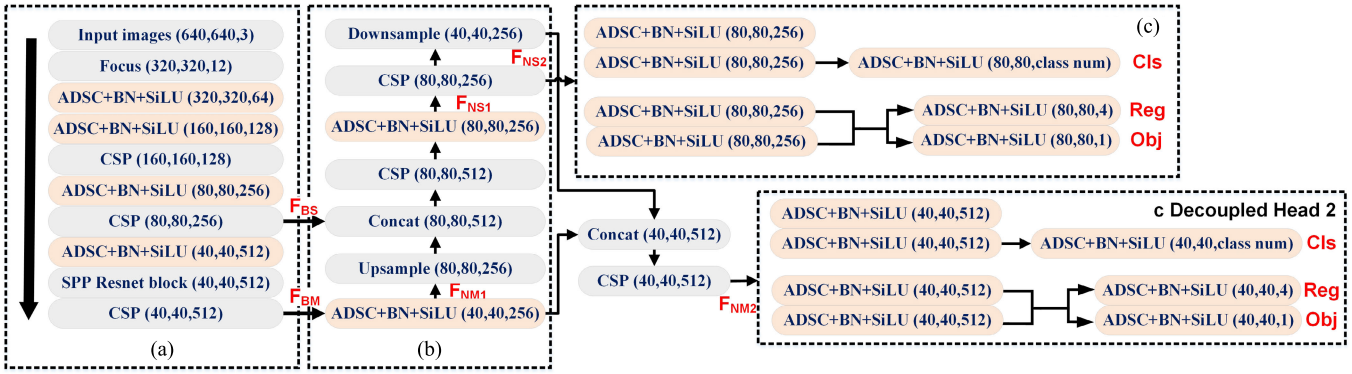


Fig. 14. Simplified network parameters and structure for decoupled head detection of vehicle targets.

target location is derived, and this information is used as input to the detection algorithm.

However, relying solely on the location information does not yield precise target detection results. Therefore, in Section III-C, a complete and improved lightweight detection algorithm is proposed. This algorithm, based on the location information, further detects other abstract features of small vehicle targets. Specifically, the localization results can help the detection algorithm narrow the search range or offer regions where the target may appear. This means that the detection algorithm no longer needs to search the entire space or a large region but can instead focus on the smaller area supplied by the localization algorithm. This collaborative relationship improves the detection efficiency to some extent and reduces the false detection rate. In other words, the location information offers the detection algorithm additional target features, enhancing its accuracy.

Moreover, the accuracy of the localization results directly influences the performance of the detection algorithm. If the localization error is significant, the detection algorithm may need to expand its search area, which could lead to a higher false detection rate and increased CC. Conversely, accurate localization results can significantly improve the success rate of detection. Therefore, the localization results not only provide crucial information for the detection algorithm but also directly affect the accuracy and efficiency of subsequent detection steps.

The critical point is that the location information provided by the localization method significantly impacts the accuracy of the detection algorithm. The relevant experimental results are detailed in Section IV, particularly in Section IV-D.

TABLE IV
NUMBER OF SAR-AIRCRAFT-1.0 DATA DIVIDED TRAIN AND TEST

Class	Train	Test	Total
A330	278	31	309
A320/321	1719	52	1771
A220	327	460	3730
ARJ21	825	362	1187
Boeing737	2007	550	2557
Boeing787	2191	454	2645
Other	3223	1041	4264
Total	13513	2950	16463

TABLE V
NUMBER OF MSAR-1.0

Class	Bridge	Ship	Oil Tank	Aircraft
Number	1851	39858	12319	6368

TABLE VI
NUMBER OF FAST-VEHICLE

Monostatic			Bistatic	
Month	Depression	Number	Flight	Number
March	15°	151	1	289
	31°	194	2	299
	45°	169	3	294
Total	-	514	4	287
July	26°	231	5	292
	31°	286	6	288
	37°	297	7	299
	45°	295	8	283
Total	-	1109	9	292

IV. EXPERIMENTS AND ANALYSIS

A. Experimental Data

To verify the scalability of our approach, two common SAR target detection datasets were employed in this study, including ground aircraft targets and near-coastal ship targets. The ground aircraft target dataset (SAR-AIRcraft-1.0), is derived from the Gaofen-3 satellite [41], and the main scenes come from the image data of seven aircraft types from three civil airports including Shanghai Hongqiao Airport, Beijing Capital Airport, and Taiwan Taoyuan Airport. The sample categories are shown in Table IV. In addition, the near-shore ship target dataset is sourced from the satellite-borne SAR dataset HISEA-1 (MSAR-1.0) [42]. Notably, this dataset encompasses not only ship targets but also oil tanks, bridges, and aircraft. The class quantities are detailed in Table V. Table VI presents the categories and quantities of FAST-Vehicle data utilized in our experiments. It is worth noting that, to mitigate potential overfitting resulting from fixed target positions, we have implemented mobility and adjustments to the targets within the panoramic view.

B. Experimental Setup

All the work was carried out on a server. The hardware includes an Nvidia Quadro RTX6000 GPU, 2400-MHz

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TABLE VII
CONFIGURATION OF LTY-NETWORK HYPERPARAMETERS

Init learning Rate	Min learning Rate	Decay type	Batch
0.001	0.00001	Cosine Annealing	24
Epoch	Optimizer	Weight Decay	Momentum
200	Adam	0.0005	0.937

memory, and an Intel Xeon Silver 4210 processor with 128 GB of RAM. The software comprises CUDA 11.1, Python 3.7, and Pytorch 1.9.0, along with PyCharm (64-bit Windows) as the integrated code management environment based on the Python IDE. In addition, the network hyperparameters used in the experiments are detailed in Table VII.

To validate the ability of the proposed method for detecting small SAR targets in ultracomplex scenarios, six sets of experiments were designed in this study (see Table VIII).

EXP 1 involves training and testing the algorithm using monostatic data with different depression angles collected at the same time. This is to validate the detection ability of the proposed method in response to the base data exclusively from the same source.

EXP 2 focuses on utilizing cross-month data for training and testing. The SAR sensor data for July has been modified with an improved antenna amplifier from the March data, resulting in some similarities and differences between the two datasets. This experiment aims to verify the detection capability of the algorithm when applied to an extended dataset.

EXP 3 is training and testing the algorithm via bistatic data from different flight missions. This experimental setup is designed to assess the detection capability of the method by fundamental bistatic SAR images.

EXP 4 is to evaluate the transferability of the algorithm. This experiment is more challenging compared to the previous three groups due to the differing data acquisition methods for monostatic and bistatic data.

EXP 5 serves as a comparative experiment. While the proposed method was originally designed for ground-based vehicle small targets, publicly available high-resolution ship datasets are used for validation.

EXP 6 incorporates more attention to SAR aircraft target detection by publicly aircraft datasets for comparative validation.

C. Experimental Results

According to EXP 1–EXP 6 in Section IV-B, the proposed method has been verified. To comprehensively evaluate the accuracy and effectiveness of this algorithm, four metrics were used for validation. The specific metrics are as follows: recall [see (20)], mean average precision 0.5 (mAP 0.5) and mAP 0.75 [see (22)], and $F1_{\text{score}}$ [see (23)]. Moreover, the accuracy of individual sample classification is also calculated in this experiment to analyze the experimental results in detail

$$R = \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$P = \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

$$\text{mAP(IoU)} = \int_0^1 P(R)dR, \quad \text{IoU} = [0.5, 0.75] \tag{22}$$

$$F1_{\text{score}} = 2 \times \frac{P \times R}{P + R}. \tag{23}$$

Here, $P(R)$ represents the integration curve of P. In addition, TP denotes the number of positive samples correctly predicted as positive by the classifier, TN denotes the number of positive samples incorrectly predicted as negative by the classifier, FP denotes the number of negative samples incorrectly predicted as positive by the classifier, and FN denotes the number of negative samples correctly predicted as negative by the classifier.

1) Accuracy of LTY-Network Based on Different Datasets: We analyze each experiment according to the experimental results in Table IX.

EXP 1: On the one hand, the model achieved excellent performance with a high mAP 0.5, indicating its effectiveness in detecting SAR vehicle small targets in homogeneous monostatic SAR images. On the other hand, the mAP 0.75 showed a slight decrease, suggesting a decrease in accuracy at higher IoU thresholds. Nevertheless, both the recall and $F1_{\text{score}}$ remained at high levels, demonstrating good overall performance. EXP 1–2 showed similar performance to EXP 1–1 in terms of mAP 0.5, but it exhibited a significant drop in mAP 0.75, suggesting poorer performance at higher thresholds. The recall and $F1_{\text{score}}$ were also lower than EXP 1–1, illustrating slightly inferior overall performance. We attribute this is since the quality of the SAR March dataset is overall lower than the quality of the July data, with a large number of vehicle small targets missing and deformed. In addition, the March dataset had significantly fewer samples compared to the July dataset, which could have diminished the generalization ability of the model and affected the detection accuracy. In summary, our proposed method is applicable for SAR vehicle target detection in ultracomplex scenarios.

EXP 2: The model demonstrated moderate performance in cross-month monostatic SAR image small target detection, as indicated by the mAP 0.5 and mAP 0.75. This is influenced by the fact that the two imaging conditions are distinct, in other words, the image data impacts the target detection accuracy. However, both the recall and $F1_{\text{score}}$ were quite high, with the $F1_{\text{score}}$ reaching 81.33%, meaning a good balance between accuracy and recall. From an application perspective, the proposed approach can handle the detection of SAR ground small vehicle targets in ultracomplex environments.

EXP 3: Despite the experimental results of EXP 3–1 being the lowest in EXP 3, all metrics were well balanced, with mAP 0.5 and mAP 0.75 being relatively close values, which shows a stable performance across different thresholds. EXP 3–2 had higher scores than EXP 3–1, particularly with an $F1_{\text{score}}$ of 84.18%, indicating improved model performance. EXP 3–3 has the most outstanding results, with high scores in all indicators, especially the recall rate exceeding 90% and the $F1_{\text{score}}$ approaching 90%, which illustrates the effectiveness of this model. EXP 3–4 showed metrics similar to EXP 2, implying good model robustness. EXP 3–5 had a large discrepancy

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TABLE VIII
TRAIN AND TEST DIVISION FOR SIX SETS OF EXPERIMENTS

EXP ID		Train		Test	
		Dataset	Number	Dataset	Number
EXP 1	EXP 1-1	Monostatic July 26°, 31°, 37°	814	Monostatic July 45°	295
	EXP 1-2	Monostatic March 15°, 31°	814	Monostatic March 45°	169
EXP 2		Monostatic July all	1109	Monostatic March all	514
EXP 3	EXP 3-1	Bistatic 1	289	Bistatic 2	299
	EXP 3-2	Bistatic 3	294	Bistatic 4	287
	EXP 3-3	Bistatic 5	292	Bistatic 6	287
	EXP 3-4	Bistatic 7	292	Bistatic 8	288
	EXP 3-5	Bistatic 8	299	Bistatic 9	283
EXP 4		Monostatic July / March all	1623	Bistatic all	2623
EXP 5		MSAR-1.0 (70%)	19914	MSAR-1.0 (70%)	8534
EXP 6		SAR-AIRcraft-1.0 (82%)	13513	SAR-AIRcraft-1.0 (18%)	2950

TABLE IX
EXPERIMENTAL RESULTS FOR EXP 1–EXP 6

EXP ID		Exprimental Results			
		mAP 0.5	mAP 0.75	Recall	F1-Score
EXP 1	EXP 1-1	91.32%	75.23%	85.15%	82.22%
	EXP 1-2	91.13%	69.98%	84.40%	81.01%
EXP 2		82.19%	63.43%	76.22%	81.33%
EXP 3	EXP 3-1	82.53%	65.24%	77.82%	80.64%
	EXP 3-2	84.70%	70.39%	81.64%	84.18%
	EXP 3-3	90.65%	76.33%	91.07%	88.91%
	EXP 3-4	85.83%	68.35%	82.87%	84.63%
	EXP 3-5	83.91%	67.47%	79.48%	81.18%
EXP 4		67.91 %	44.20%	61.82%	68.32%
EXP 5		83.27%	45.21%	77.25%	79.81%
EXP 6		92.82%	65.80%	91.27%	87.57%

between mAP 0.5/mAP 0.75, but still exhibited high recall and $F1_{\text{score}}$, indicating potential for improvement at higher IoU thresholds. EXP 3 suggests that the method of the article exhibits greater stability on bistatic SAR images compared to monostatic SAR images, although detection accuracy is slightly inferior. We attribute this to the variations in the azimuth and depression angles of the receiver and transmitter between each training and testing data.

EXP 4: This represents the most challenging experiment among the six groups. It involved not only different-source SAR target imaging but also different imaging modes. Although EXP 4 achieved lower scores in all metrics compared to other experiments, especially with an mAP 0.75 of only 44.20%, a detection accuracy of 67.91% still illustrates the reference value of our method in detecting SAR small vehicle targets under ultracomplex environments, diverse imaging conditions, and different imaging modes.

EXP 5: This was an extended experiment based on the MSAR 1.0 dataset for ship detection. Its mAP 0.5 outperforms existing algorithms [43] and its model size is far smaller than existing methods [44], but the mAP 0.75 drops obviously, indicating that the performance of the model is insufficient at higher thresholds. Nonetheless, both the recall and $F1_{\text{score}}$ were satisfactory, suggesting an overall acceptable model performance. This experiment demonstrates the marked scalability of our method, which can be applied not only to SAR ground small target detection but also to the detection of targets such as ships and bridges on the sea surface.

EXP 6: This produced unexpected results. The model exhibited excellent performance with a quite high mAP 0.5 and recall at low IoU thresholds. By contrast, there was a decrease in mAP 0.75, showing that improvement is needed at high thresholds. We will further provide explanations for the superior performance of our model on SAR aircraft datasets in Section IV-C2.

In conclusion, our model demonstrates excellent performance in both monostatic SAR images and bistatic SAR images under complex environments. The detection accuracy can exceed 90%, accompanied by high recall and $F1_{\text{score}}$, indicating stable performance. Furthermore, our proposed method exhibits substantial scalability and can be applied to the detection of SAR ground small targets, ships, bridges, and other diverse targets.

2) Detection Accuracy Analysis of Individual Targets: To further explore the detection accuracy of the model, experiments on the accuracy of each target category were also conducted in this project. Considering the reduction of repetition workload, we only adopted one of the subexperiments of EXP 1 and EXP 3 for analysis. The experimental results are shown in Fig. 15.

In EXP 1–1, the detection accuracy exceeds 90% in most categories, consistent with the results in Table IX. However, 62LT, 63APTV, 63AT, and 63CAAV fall below this threshold. From Fig. 16, it is evident that these targets are closer in distance and exhibit higher similarity. Therefore, we infer that these factors contribute to the lower detection accuracy observed in these categories.

Results from EXP 2 indicate metrics inferior to EXP 1, with individual target detection accuracy generally lower as well. This is because the monostatic March data and the monostatic July data vary in terms of image size, surrounding environments of the targets, and SAR sensor amplification characteristics. These factors collectively contribute to the decreased accuracy in SAR target detection.

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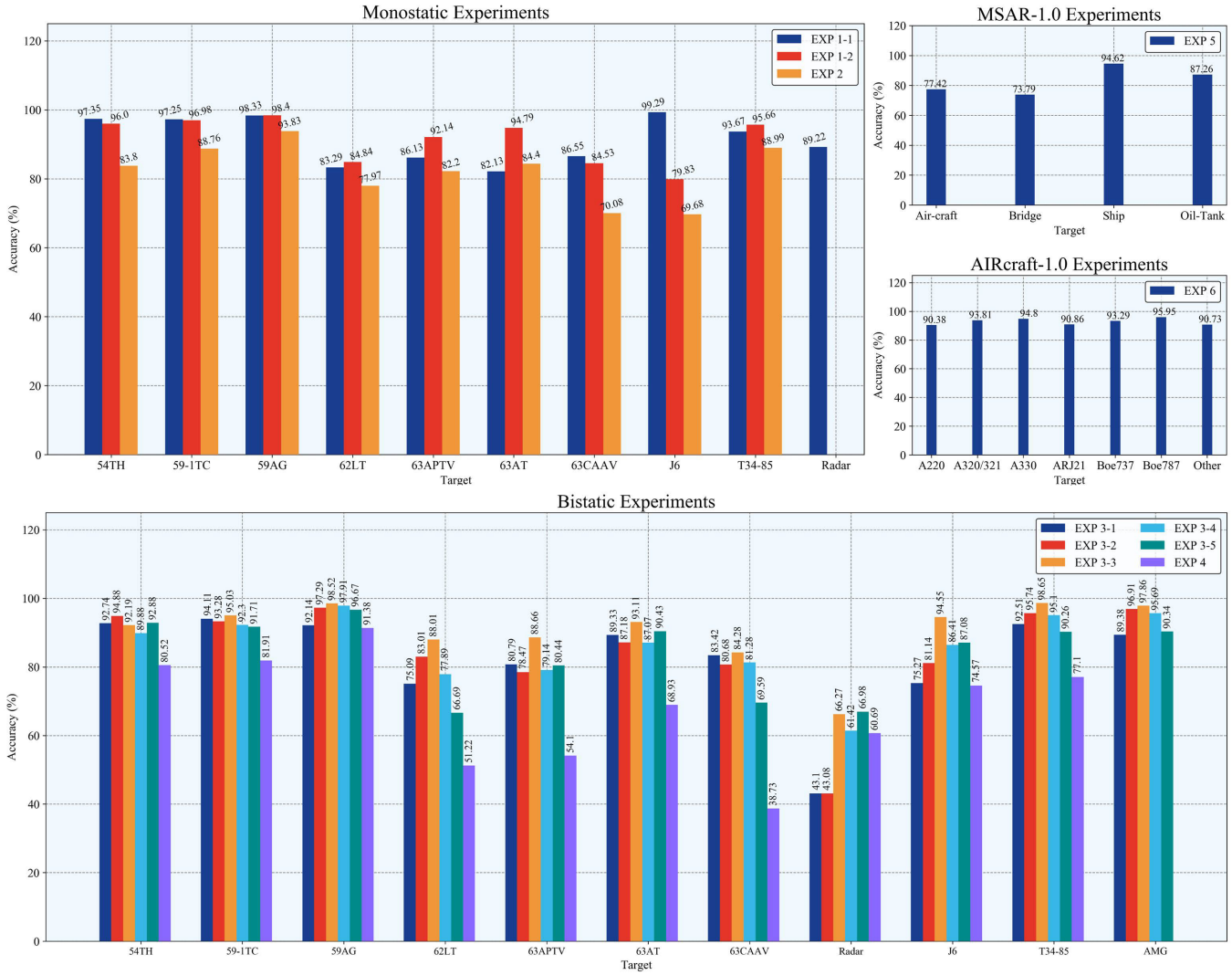


Fig. 15. Detection accuracy for individual targets in EXP 1–EXP 6.

What is noteworthy in the EXP 3 bistatic SAR experiment is that radar target detection accuracy is notably lower compared to other target classes. Fig. 16 (EXP 3–1 and EXP 3–2) reveals that radar targets suffer from reduced scattering intensity and are prone to missing due to their placement at the edge of the imaging scene. Moreover, observations indicate that the radar operates in a rotating mode rather than being stationary, leading to unstable target morphology in the images. The limited sample number and unstable target attitude further contribute to the significantly lower detection accuracy of radar targets compared to others.

EXP 4 presents a more complex scenario than EXP 2, with training and testing data comprising both monostatic and bistatic SAR datasets. The substantial differences result in markedly lower individual target detection accuracy, particularly evident in categories 62LT, 63APTV, 63AT, and 63CAAV. These categories not only share similar appearances (consistent with monostatic data) but also suffer from interference from scene-strong scattering blocks, which cause target absence and deformation.

The results of EXP 5 show that our method effectively retrieves target positional information, for extremely small vessels [see Fig. 16 EXP 5(c)], and the oil tank targets on the false alarm also exist. Nevertheless, depending on the ability of the detection algorithm to detect small targets, it is also able to detect every target.

EXP 6 shows high detection accuracy across all target categories. Fig. 16 illustrates that aircraft predominantly reside in areas with weaker reflection zones such as parking lots, minimizing ground clutter interference. Furthermore, variations in each morphology of aircraft and size enhance target distinctiveness, thereby improving detection accuracy.

D. Ablation Experiments

1) Influence of Target Location Information on Accuracy: To further explore the reasons for improving detection capabilities in ultracomplex scenes, we conducted six sets of ablation experiments based on the experiments in Section IV-C2. The results are presented in Fig. 17.

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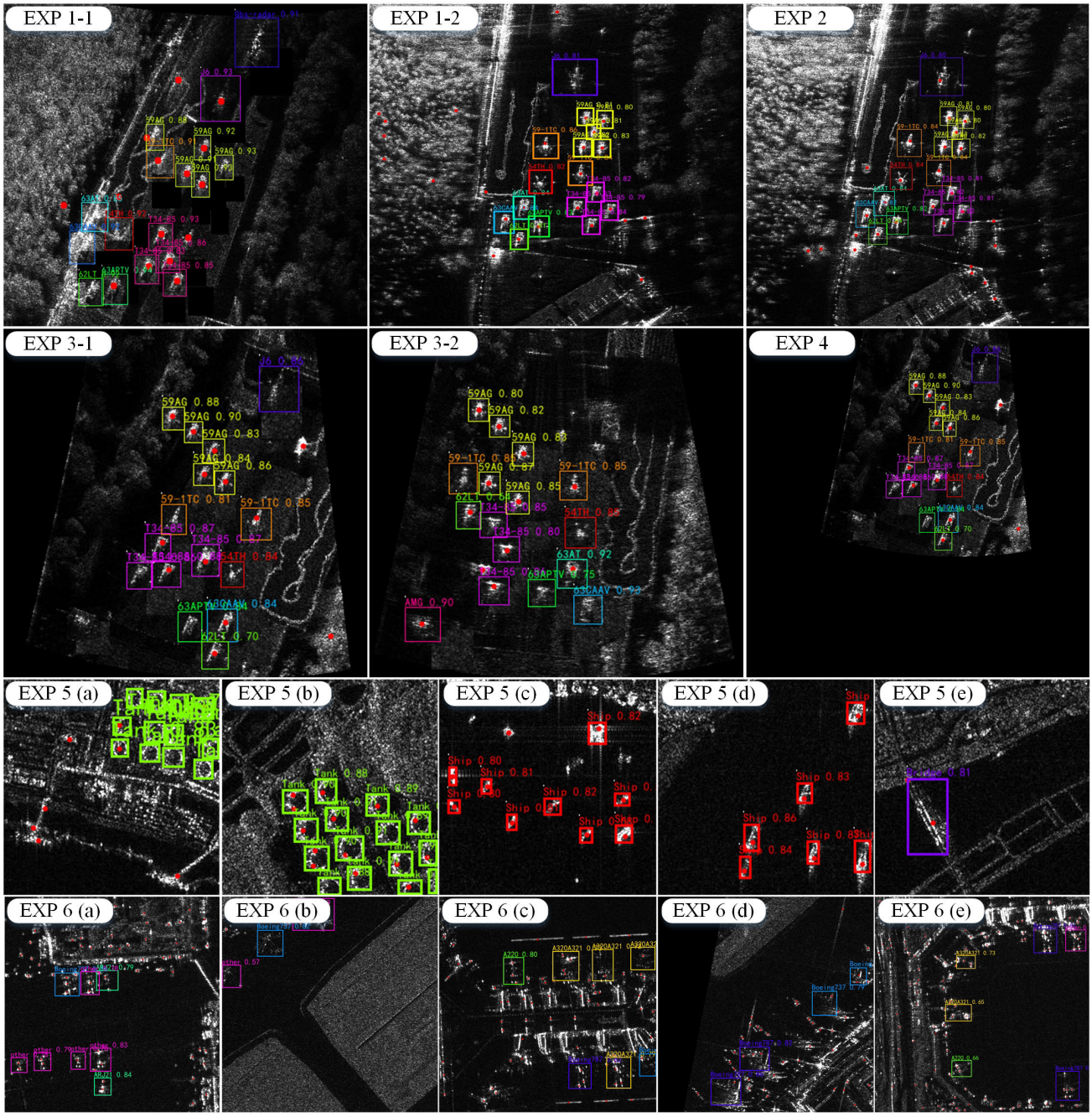


Fig. 16. Target detection results in EXP 1–EXP 6.

From the experimental results, it is found that most of the experimental groups, including target location information resulted in superior performance compared to scenarios where such information was not included. This finding suggests a significant role of target location information in enhancing detection accuracy. This reveals that the location information of the target has a remarkable impact on enhancing the detection accuracy.

Specifically, the presence of target location information on our FAST-Vehicle dataset is the ability to improve approximately more than 2%, for instance, EXP 1 and EXP 2. In particular, the target location information becomes more critical when the detection accuracy of the model is below 80%, it contributed to an enhancement of detection accuracy by approximately 8%–10%, such as EXP 3 and EXP 4. This indicates that the essentiality of location information is amplified when the detection accuracy of the model is low, whereas its impact diminishes when the accuracy is high. This phenomenon can also be derived from the extended experiments EXP 5 and EXP 6. Furthermore, we noted that in the extended experiments, the improvement in detection accuracy due to location information was not pronounced, amounting to only about 1%–1.5%. We attribute this to the relatively fewer background interferences in the extended dataset compared to FAST-Vehicle, resulting in less discernible effects.

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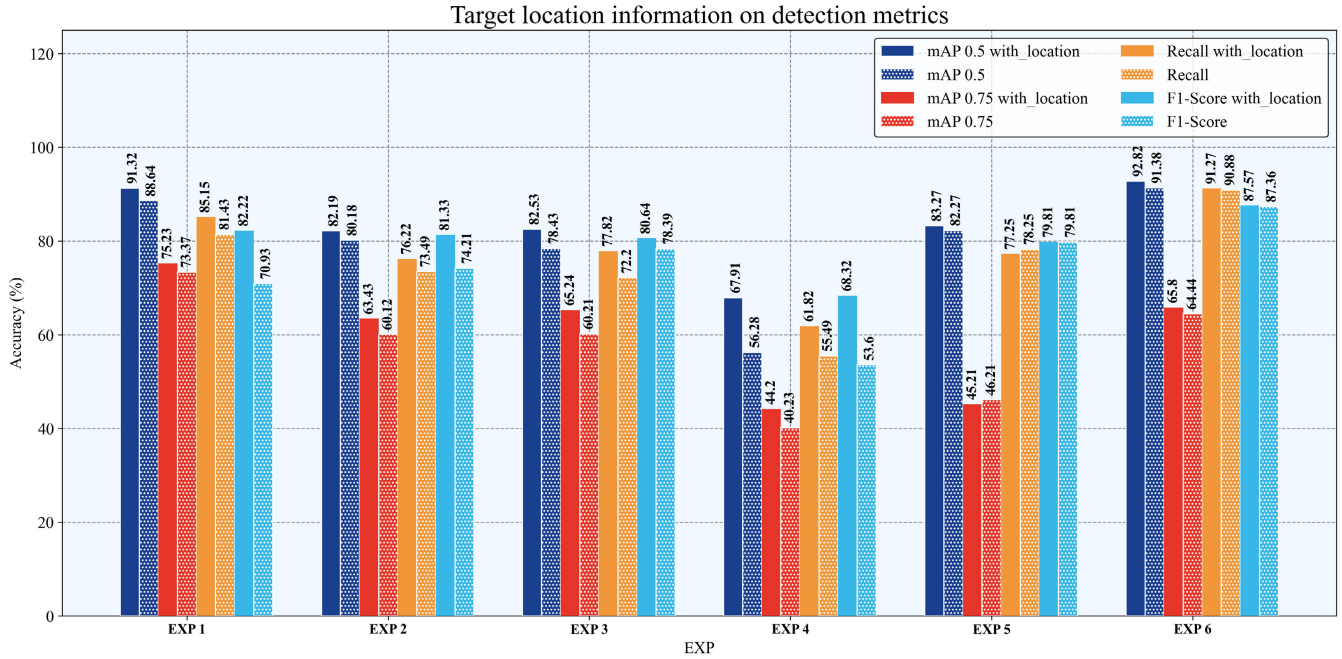


Fig. 17. Experimental results of metrics for detection with and without target position information.

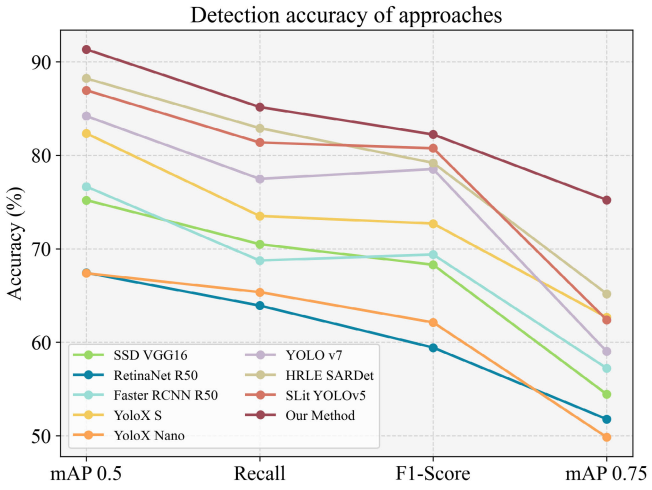


Fig. 18. Experimental results of metrics for detection with and without target position information.

In summary, the analysis of the results from the six experiments indicates the significant function of target location information in improving detection accuracy. Particularly, in terms of mAP 0.5 and the recall metrics, models incorporating location information generally outperformed those without it. In some instances, models with target location information may exhibit slightly inferior performance in the mAP 0.75 metric.

2) Comparison of Accuracy and Speed With Other SAR Detection Methods: To validate the proposed the capability of method in addressing the speed challenges of SAR target detection, our research team conducted an algorithmic speed verification based on six sets of experiments. The metrics include frames per second (FPS) and floating point operations per second (FLOPS). FPS represents the number of images the model can process per second, serving as a critical measure of detection speed. A higher FPS indicates better performance of the detection model. FLOPS measures the number of floating-point operations per second. On the same hardware, a network requiring fewer GFLOPS suggests that it can handle more images, thus indicating a faster model. Our comparative experimental results are illustrated in Fig. 18.

Certainly, a comprehensive experiment necessitates comparisons with other existing classical and outstanding algorithms. We selected several well-known detection algorithms for comparison, including one-step detection algorithm, SSD [23], RetinaNet-R50 [45], YoloX-S and YoloX-Nano [37], YOLO v7 [14], and two-step detection algorithm Faster-RCNN-R50 [24]. In addition, we included excellent SOTA algorithms developed for SAR target detection, including HRLE-SARDet [44] for accuracy advantage and SLit-YOLOv5 [46] for speed advantage. The article used EXP 1–1 as the baseline for reference, with the experimental results presented in Table X.

On the one hand, our method demonstrates outstanding performance in terms of accuracy, achieving the highest mAP 0.5 (91.32%) and mAP 0.75 (75.23%). This indicates that our approach can detect objects with greater precision across various IoU thresholds. Furthermore, both the $F1$-score and recall are comparatively high, reflecting a well-balanced performance in object detection.

On the other hand, in terms of speed, our method exhibits a relatively low parameter count (1.98M) compared to some models with larger parameter sizes. This suggests that our approach maintains high accuracy while ensuring reduced model complexity. In other words, a smaller model can lead to lower FLOPS. Table X shows that the FPS (25) and FLOPS

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TABLE X
PERFORMANCE COMPARISON OF VARIOUS METHODS

Method	Accuracy				Speed			
	mAP 0.5	Recall	F1-Score	mAP 0.75	Parameters (M)	FPS	FLOPS (G)	Second (S)
SSD VGG16	75.20%	70.49%	68.28%	54.45%	35.64	1.05	34.85	0.95
RetinaNet-R50	67.43%	63.92%	59.43%	51.78%	33.01	0.22	151.54	4.35
Faster-RCNN-R50	76.65%	68.74%	69.38%	57.23%	41.75	0.23	134.38	4.42
YoloX-S	82.34%	73.50%	72.69%	62.67%	8.94	1.23	26.78	0.81
YoloX-Nano	67.38%	65.36%	62.12%	49.85%	0.91	29.44	1.08	0.03
YOLO v7	84.19%	77.47%	78.52%	59.02%	36.93	0.31	104.7	1.35
HRLE-SARDet	88.21%	82.90%	79.18%	65.19%	1.09	13.22	2.50	0.07
SLit-YOLOv5	85.94%	81.36%	80.74%	62.40%	**0.30**	**39.74**	**0.80**	**0.02**
Our Method	**91.32%**	**85.15%**	**82.22%**	**75.23%**	1.98	25.00	1.41	0.04

(14.09 G) of our model are slightly inferior to some other models, such as YoloX-Nano and SLit-YOLOv5. This means that under the same hardware conditions, our method probably not be able to achieve the same real-time performance as these models. Nevertheless, the difference is marginal, around 0.01–0.02, which can be considered negligible.

On the whole, our method possesses significant advantages in accuracy but is slightly less competitive in terms of speed compared to certain models. As a result, our approach remains relatively lightweight and can still achieve high FPS, making it a very viable option in resource-constrained environments.

E. Discussion

Current research in SAR target detection is primarily focused on maritime vessels, due in large part to the abundant availability of SAR ship datasets. In contrast, research on the detection of small ground vehicle targets has been constrained by limited data and has thus focused mainly on the MSTAR dataset. Moreover, ground environments are typically more complex compared to open water environments where targets on the surface of the water are located. The primary contribution of this article is the use of the inherent features of SAR images combined with a lightweight detection algorithm to enhance detection accuracy while maintaining fast detection speeds without significant tradeoffs.

To be frank, there are certain limitations in this study. For instance, although our approach achieves high accuracy on monostatic SAR data, the recognition accuracy for bistatic SAR data remains below 90%. This issue warrants further research on bistatic datasets. We think that this phenomenon is mainly due to the fact that the receiver in bistatic SAR systems does not capture the reflected electromagnetic wave intensity as effectively as the monostatic systems where the flight trajectory is perpendicular to the target. This results in unstable imaging conditions. Last but not least, the image resolution of bistatic SAR is typically lower than that of monostatic SAR.

Compared to conventional SAR target detection research, the most significant contribution of this study is the focus on bistatic SAR systems. The research on bistatic SAR systems in this article is meaningful in three key aspects: it validates the ability of the algorithm to detect small vehicle targets in ultracomplex environments, demonstrates the robustness of the proposed method across both monostatic and bistatic SAR systems, and serves as a foundational exploration for future swarm-based SAR systems. In future work, we aim to improve the detection accuracy of small ground vehicle targets in bistatic SAR data and advance the practical application of the proposed method. Furthermore, we will pay more attention to the development of SAR swarm systems, providing more possibilities for actual military applications.

V. CONCLUSION

At present, due to complex scenarios and inferior real-time performance of detection algorithms, several SAR target detection methods exhibit limited practicality in detecting small targets in ultracomplex scenes. To address this issue, this article proposes an effective and fast one-step SAR small target detection algorithm, named LTY-Network. Validation experiments were conducted on publicly available datasets of ships and aircraft, as well as on monostatic and bistatic SAR datasets developed by our lab. Experimental results demonstrate that the proposed algorithm achieves the highest detection accuracies of 91.32% and 90.65% on monostatic and bistatic SAR datasets, respectively. Furthermore, it attains 92.82% on an extended aircraft dataset. Ablation and comparative experiments confirm the effectiveness of our method, which operates at 25 FPS, approaching the smooth frame rate of visual perception. This further validates the efficacy of the proposed method in detecting SAR small targets in complex environments, enhancing its practical value in both military and civilian applications. Future work will explore SAR target recognition and detection techniques for swarm UAVs, aiming to broaden the application scope of SAR images. This approach seeks to obtain more comprehensive target information from a saturated detection perspective, thereby further improving detection and recognition accuracy.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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